

FIN 7560 – Valuation and Growth

FINAL EXAM

Instructions:

1. This is an open-book, open-notes exam.
2. There are 70 total points; points per question are indicated below.
3. Insert your answers (either written or typed), and upload the completed exam to Canvas before the end of the day, February 8, 2026.
4. Good luck!

1. (5 points) Suppose you are trying to determine whether your company should go forward with producing and selling a new product that your company has developed. As part of this valuation, you are trying to identify the incremental free cash flows that will occur if your company goes forward with production. Which of the below are incremental free cash flows that should be included in your valuation?
 - I. Expenses from past R&D projects that helped develop this product
 - II. The future use of excess capacity that would otherwise go unused
 - III. Expected erosion of sales that will occur in your other, competing product lines because of entry by other competitors into this new product space
 - a. I only
 - b. II only
 - c. I and II
 - d. II and III
 - e. All of the above
 - f. None of the above
2. (5 points) MGM operates in two different industries: casinos and movies. Its casino operations have very volatile (i.e., risky) cash flows, whereas its movies operations have less volatile cash flows. If the company attempts to use its weighted average cost of capital for the entire firm to evaluate projects in each division, it will...
 - a. Accept too many projects for both divisions
 - b. Reject too many projects for both divisions
 - c. Accept too many projects in its movies division, and reject too many projects for its casino division
 - d. Accept too many projects for its casino division, and reject too many projects for its movie division
 - e. It's unclear; we need more information to answer this question.
 - f. None of the above
3. (5 points) Eleanor is valuing a young company using a multiples approach. To do this, she has identified five comparable firms, but none of them are great matches for the company she is valuing, though some are better matches than others. Using this information, how could she improve her valuation?
 - a. She should calculate the multiple for each of the five firms and use the average of this so as to reduce the noise of the estimated value she gets
 - b. She should only use the best one or two comparables and ignore the others
 - c. She should calculate the multiple for each of the five firms and use the median of this so as to reduce the influence of the less comparable firms
 - d. It's unclear because there is no correct answer here... any of the above are reasonable things to do when doing a multiples valuation

4. Acme, Inc. has 2 million outstanding common shares that sell at price \$25 per share. The common shares are expected to have a dividend next year of \$3 and the dividends are expected to grow at a rate of 3 percent a year thereafter. The company has \$40 million par value of debt that is priced 95 percent of par and has a yield to maturity of 10 percent. The corporate tax rate is 35%. Use this information to solve the below questions.

a. (10 points) What is Acme's cost of debt? What is its market value of debt?

b. (5 points) Using the above information, what is Acme's cost of equity?

[Hint: You should use the dividend growth model to find the cost of equity].

c. (10 points) If Acme is planning to keep its future mixture of debt and equity financing the same, what would be an appropriate cost of capital for Acme, Inc. to use if it were trying to calculate the NPV of a project of similar risk to its existing operations?

[Hint: You will need to use a combination of your answers from parts (a) and (b). If you are unable to solve either part (a) or (b), just make a guess and use that to solve part (c)]

5. Google is thinking of acquiring VRX, a new virtual reality software company, and Google is attempting to determine the highest price it should be willing to pay for VRX. It expects VRX's FCF will equal \$20 million next year (i.e. $t = 1$), \$25 million the year after (i.e. $t = 2$). After that year, however, Google is uncertain what future FCF will be. However, Google notices that other, more mature companies in the VR software industry have an average market value of debt and equity to EBIT ratio of 3, and they project VRX's EBIT to be \$50 million in year $t = 2$. VRX has \$2 million in excess cash on hand now, \$15 million in debt, and 10 million outstanding shares. VRX's WACC is 20 percent.

- a. (5 points) Using a multiple, what is a possible terminal value for VRX in year 2?
- b. (10 points) Using your answer from part (a) and the present value of cash flows, what is the value of VRX's assets (i.e., total firm value)?
- c. (10 points) Google doesn't believe there is any scope for synergies from the acquisition of VRX. Given that and using your answer from part (b), what is the most that Google should be willing to pay per share to acquire VRX?

- d. (5 points) Suppose Google is acquiring VRX partly to keep the company from being acquired by Apple because it fears that Apple will use VRX's software in its future iPhones, further hurting Google's struggling cell phone business. If true, how might this affect the price that Google is willing to pay, and why? *[2-3 sentences is fine]*